An Analysis of Lobbying by Cities

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1. Introduction

Although lobbying is typically associated with private interest groups, such as corporations or private citizen organizations, a nontrivial amount of lobbying is conducted by public interest groups as well. Among these public interest groups are individual cities, who are active lobbyists of the federal government. In fact, over 60% of US residents live in cities that have hired a lobbyist at some point to advocate for local interests at higher levels of government (Payson 2021, 1). These cities lobby in order to procure earmarks, grants, or other resources, as well as influence federal regulations that affect local municipalities (Payson 2021, 25).

In this paper, I extend the analysis from Goldstein and You (2017), who use cities' lobbying activity data from the Lobbying Disclosure Act of 1995. Under the Act, any interest group is required to publicly disclose when lobbying has occurred, for what purpose, and how much was spent, among other information (Office of the Clerk, U.S. House of Representatives 2021). The original researchers hypothesize and find that the probability of lobbying increases with the city-state public goods gap, which is when cities have different preferences for public spending from their states (Goldstein and You 2017). I fit two kinds of models: linear probability models (LPMs) and logistic regressions to explore and extend their findings on what factors predict a city's propensity to lobby the federal government. I also add and examine the effect of original indicators for federal institutions, including public universities and federally funded research facilities. Finally, I run diagnostics on my models and discuss the limitations of the data.

2. Data and Research Question

The data spans thirteen years from 1999-2012 and contains yearly observations on 1,262 cities with populations of at least 25,000 people. Cities enter the dataset if they meet this population threshold and if there is adequate information on them available through the Census of Governments, regardless of whether or not they lobby. Of these, 541 cities (24%) submitted at least one lobbying report and 721 cities did not submit any.

Additional figures in the appendix show the most active lobbyist cities by expenditure, as well as the top issues of concern highlighted in the reports. Somewhat unsurprisingly, the top issue of concern for cities is the budget, for which over eight thousand reports were submitted. The city that spent the most on lobbying was New Orleans, Louisiana, which spent over $7.3 million over the course of the study.

The original researchers focus on when the public spending preferences of a city differ from those of the state, thereby creating an incentive to lobby the federal government for resources instead (Goldstein and You 2017). I extend their analysis about the relevance of public goods and services by including original indicators for federal facilities, including research facilities and public R1 universities. I hypothesize that cities that are the locations of major, federally funded institutions are more exposed to the federal government's decisions regarding those institutions, especially if they comprise significant portions of a city's economy or employment opportunities. As a result, these cities may be more inclined to lobby the federal government for their desired outcomes regarding these institutions.

Table 1: Descriptive statistics for variables.

Variable	N	Mean	Std. Dev.	Min	Pctl. 25	Pctl. 75	Max
Lobbied	17668	0.24	0.43	0	0	0	1
Spending on Lobbying	17668	18284	46550	0	0	0	635000
Log of City Population	17668	11	0.78	9.8	10	11	16
Log of City Revenue	17668	12	1.1	7.4	11	12	18
Research Facility	17668	0.022	0.15	0	0	0	1
Public R1 University	17668	0.093	0.29	0	0	0	1
Intergovernmental Revenue Share	17668	0.18	0.13	0	0.085	0.24	0.78
Household Poverty	17668	11	6.9	0.53	5.2	15	38
Ethnic Heterogeneity Index	17668	0.39	0.17	0.037	0.25	0.52	0.78
Income Inequality (Gini)	17668	0.42	0.053	0.27	0.39	0.46	0.64
Ideological Difference	17668	0.039	0.22	-0.62	-0.11	0.16	0.84
Direct Flight to DC	17668	0.09	0.29	0	0	0	1

I gathered data from the Carnegie Classification of Institutions of Higher Education to obtain a list of R1 doctoral universities, classified as having "very high research activity" (American Council on Education 2023). I then filtered the list down to only public universities. For research facilities, I used data from the National Center for Science and Engineering Statistics, which compiles a list of all federally funded research and development centres. Both variables were added to the data as binary indicators. Where an institution had locations in different cities, I added the indicator to both cities. For example, the University of Illinois Urbana-Champaign was counted for both Urbana and Champaign.

This paper looks at the following research question: what factors influence a city's likelihood of lobbying the federal government? I will focus specifically on city size by population, and mediate this relationship by interacting this variable with several others, including the presence of federal institutions or demographic factors like ethnic heterogeneity or income inequality.

3. Linear Probability Models

For my independent variables, I mainly examine the effect of population, log-transformed due to skew, and my original indicators of research facilities and public R1 universities. I also include in my models the following variables and controls: intergovernmental revenue share (the percentage of a city's revenue that comes from intergovernmental sources, such as state or federal government), the city-wide poverty rate, ethnic heterogeneity, income inequality, ideological difference between the city and the state, and the existence of a direct flight to DC.

I ran four LPMs of increasing complexity, the results of which are displayed in Table 2. The first model captures basic independent variables and the second includes interaction terms. The third and fourth models correct for issues I diagnosed in the first two and contain state fixed effects, as well as robust standard errors for heteroskedasticity and autocorrelation.

Beginning with Model 1, several predictors seem to be highly statistically significant: city population, revenue, poverty rate, income inequality, and the presence of a direct flight to DC all have p-values of less than 0.001. With my original variable of a research facility, it appears that the presence of a research facility increases the probability of lobbying by 8.8 percentage points, and is highly statistically significant. Comparatively, the presence of a public R1 university increases the probability of lobbying by 1.3 percentage points, but this result is not significant and should likely be disregarded for analysis. The direction of both relationships aligns with my hypothesis, as both are positive.

Model 2 extends the analysis by including interaction terms. I chose to focus on a city's population size to see if different combinations of variables affect the relationship. I then plotted the conditional marginal effects,

Table 2: Linear probability models on lobbying.

	Model 1	Model 2	Model 3	Model 4
(Intercept)	−2.161***	−3.965***		
	(0.056)	(0.409)		
Log of City Population	0.167***	0.332***	0.179***	0.179+
	(0.008)	(0.038)	(0.037)	(0.108)
Log of City Revenue	0.019***	0.275***	0.303***	0.303**
	(0.006)	(0.032)	(0.031)	(0.104)
Research Facility	0.088***	0.091***	0.073***	0.073
	(0.020)	(0.020)	(0.020)	(0.089)
Public R1 University	0.013	0.014	0.013	0.013
	(0.011)	(0.011)	(0.011)	(0.033)
Intergovernmental Revenue Share	−0.298***	−0.302***	0.012	0.012
	(0.023)	(0.023)	(0.030)	(0.078)
Poverty Rate (Household)	0.002***	−0.037***	−0.031***	−0.031+
	(0.001)	(0.008)	(0.008)	(0.018)
Ethnic Heterogeneity Index	0.152***	0.706*	−0.011	−0.011
	(0.020)	(0.322)	(0.314)	(0.797)
Income Inequality (Gini)	0.726***	−2.087+	−2.359*	−2.359
	(0.072)	(1.112)	(1.069)	(2.749)
State-City Ideological Difference	−0.066***	−0.069***	−0.009	−0.009
	(0.015)	(0.015)	(0.016)	(0.046)
Direct Flight to DC	0.068***	0.086***	0.123***	0.123**
	(0.012)	(0.013)	(0.013)	(0.042)
Log Population x Log Revenue		−0.023***	−0.020***	−0.020*
		(0.003)	(0.003)	(0.009)
Log Population x Income Inequality		0.257*	0.276**	0.276
		(0.102)	(0.098)	(0.256)
Log Population x Poverty		0.004***	0.003***	0.003+
		(0.001)	(0.001)	(0.002)
Log Population x Ethnic Heterogeneity		−0.050+	−0.005	−0.005
		(0.029)	(0.029)	(0.074)
Num.Obs.	17 668	17 668	17 668	17 668
R2	0.195	0.200	0.460	0.460
R2 Adj.	0.194	0.200	0.458	0.458
RMSE	0.38	0.38	0.36	0.36
State Fixed Effects	N	N	Y	Y
HAC Std. Errors	N	N	N	Y

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

as well as a 95% confidence interval, of population at different levels of interaction terms to investigate how each variable altered the relationship between population and lobbying. These plots are displayed in Figure 4 and discussed in the next section on results.

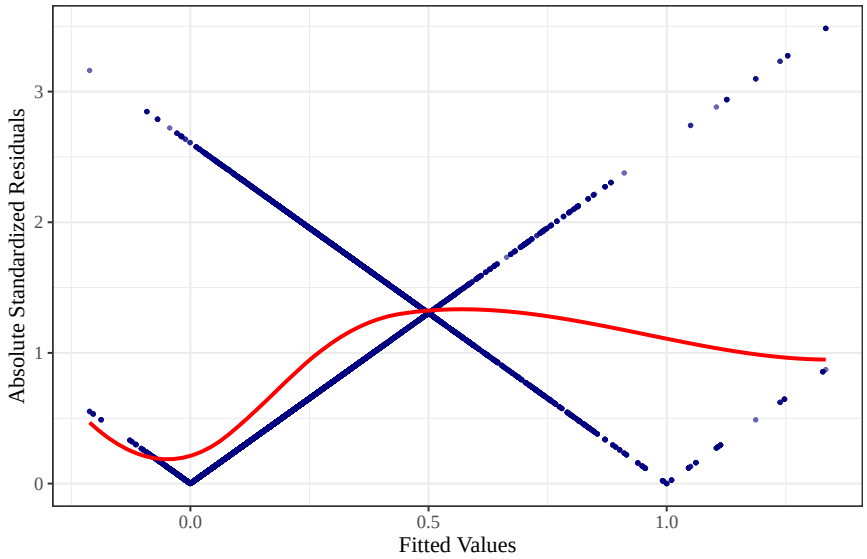
3.1 Linear Models: Diagnostics

To check the robustness of my first two models, I performed several diagnostics. I suspected the presence of heteroskedasticity, autocorrelation, or state clustering would undermine the validity of my results. I also checked for outliers. This section presents the results of several robustness checks and amendments made to the original models to account for the issues found.

To test for heteroskedasticity, I performed a Breusch-Pagan (BP) test on the first and second models. The BP test takes the null hypothesis that the error variances are constant, or homoskedastic. The results rejected the null hypothesis, indicating a highly significant presence of heteroskedasticity.

As an informal visual test, I constructed a spread-location plot, which shows the distribution of the residuals at different fitted values. Under the conditional homoskedasticity assumption, we should expect to see a straight red line indicating a constant variance across all fitted values. This is evidently not the case. Furthermore, the two distinct lines in the data reflect the binary nature of my outcome variable, and the fact that LPMs are heteroskedastic by construction.

Figure 1. Distribution of residuals across fitted values.



Non-constant error variance violates the conditional homoskedasticity assumption of ordinary least squares (OLS) regression. Although the beta estimators are still consistent and unbiased, any values constructed from the variances, including standard errors, confidence intervals, and t-tests, are no longer reliable. It becomes necessary to use standard errors that are robust to heteroskedasticity.

Another possible concern for my data is autocorrelation, which happens when observations are not independently distributed but instead are correlated over time. It is possible that a strong predictor for a city's lobbying activity in a given year is its lobbying activity in past years. For example, if cities have already established relationships with professional lobbyists, or have projects that take several years to enact, their lobbying activity may be correlated over time. I conducted a Durbin-Watson test, the results of which indicated serious positive correlation (DW = 0.46). The presence of autocorrelation causes the variance to be underestimated, creating similar problems to those of heteroskedasticity.

Finally, within-state clustering presents an issue for my LPMs. For example, if a city hears that its neighbour was able to secure funding or resources through lobbying the federal government, it may be more inclined to lobby as well. Alternatively, if a given state is very receptive to municipal interests, all cities within that state may not need to lobby the federal government. Geographical proximity therefore creates the problem of clustering, which violates the independent sampling assumption.

To adjust for all three issues identified, I applied state-level fixed effects for both Models 3 and 4, and heteroskedasticity and autocorrelation consistent (HAC) standard errors for Model 4. State fixed effects were included by adding dummies for each state into the regression, which control for the effect of state but have been suppressed in the table for concision. These should reduce clustering, isolating the effect of the independent variables on individual cities. Model 4 contains these fixed effects, as well as HAC standard errors. These do not correct for bias in my beta estimators, but they provide a consistent estimator for the variance that is robust to heteroskedasticity and autocorrelation (Newey and West 1987).

To check for outliers, I construct an influence plot for Model 1, plotting hat-values against studentized residuals. The side of each dot represents Cook's Distance, an estimate of how influential the observation is on the regression. I added labels to the observations with the highest Cook's values in an attempt to identify a substantive pattern among the outlier cities. I also plotted these cities on a map of the United States to identify any geographic patterns, such as perhaps being geographically proximate to the Capitol in Washington, D.C.

Figure 2. Influential observations plot.

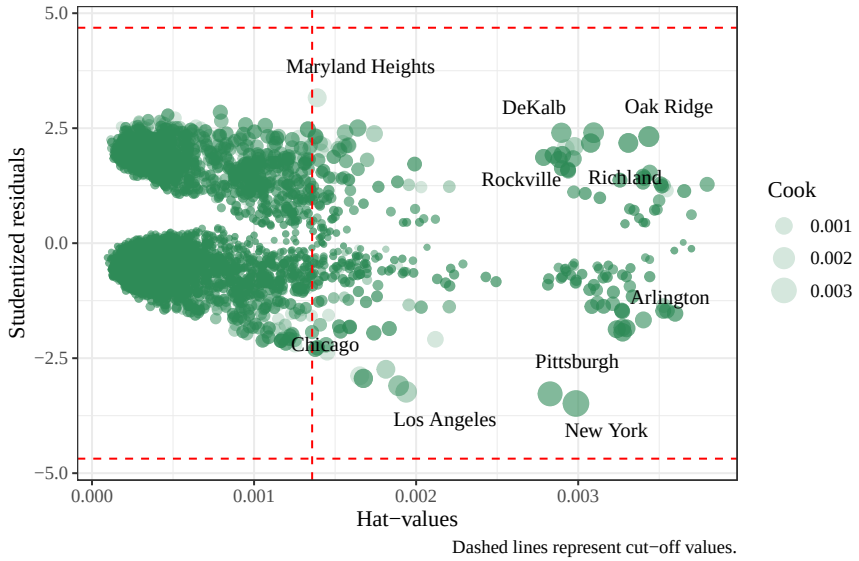


Figure 3. Map of influential outlier cities.

Some outliers are, unsurprisingly, large cities like New York City, Chicago, and Los Angeles. However, others are relatively small cities with populations only just about the sample threshold, like Arlington, Texas; Maryland Heights, Missouri; Rockville, Maryland; or Richland, Washington. From the map, there do not seem to be strong geographic patterns. I can therefore only comment on their existence but not offer a convincing reason for why they stand out. However, even the most influential outliers in my sample fall far below a Cook's Distance of 1, which would be a cause for concern. I did not make any adjustments to the model as a result.

3.2 Linear Models: Results

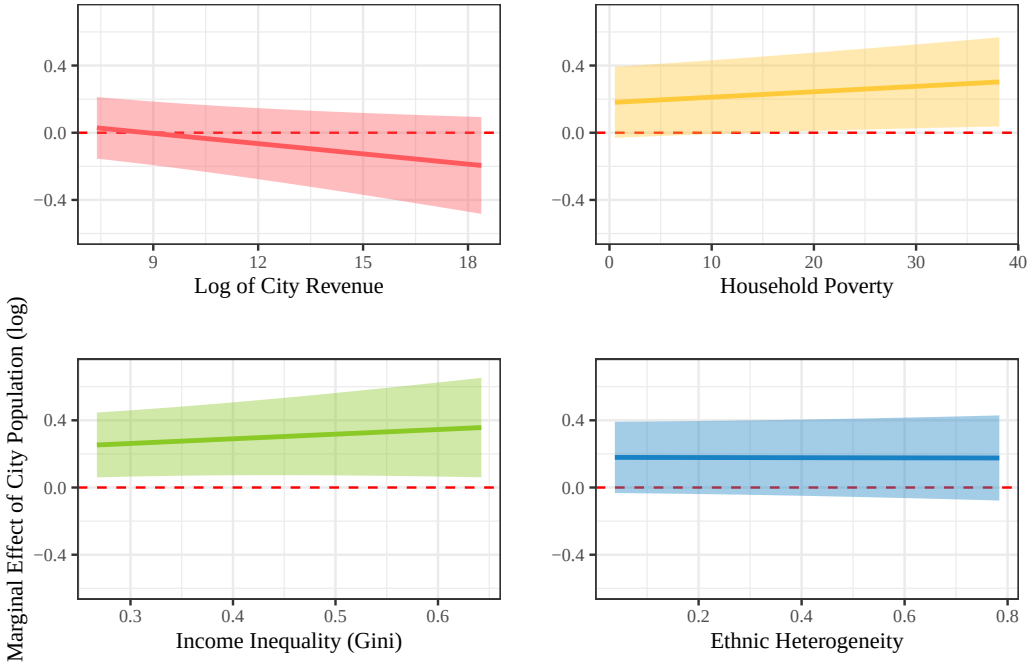
With all these corrections applied, few variables remain statistically significant. HAC standard errors are larger than their non-robust counterparts, resulting in wider confidence intervals and decreased explanatory power. By Model 4, the remaining statistically significant variables are the interaction between population and revenue at the 95% level, and the existence of a direct flight to DC at the 99% level. As the coefficient on the interaction term is negative, this means that at higher levels of revenue, population size matters less when predicting a city's likelihood of lobbying. This result may be consistent with the idea that cities primarily lobby on issues concerning public spending, and therefore feel less need to lobby when a larger population is matched with a larger pool of revenue. Comparatively, in cities with low revenue, population may matter significantly more as there are more people demanding public goods that the city must provide on a limited budget. The presence of a direct flight increases the probability of lobbying by 12.3 percentage points. This result is fairly intuitive: the cheaper and more convenient it is to reach the capital in DC, the more likely it is that a city will take the option of lobbying the federal government.

For my original indicators, the presence of a research facility increases the probability of lobbying by 7.3 percentage points. This relationship is statistically significant when the model includes state fixed effects, but not with HAC estimators as in Model 4. The presence of a public R1 university is unfortunately not significant in any model. As a result, I fail to reject my null hypothesis and conclude that the presence of a federal institution, whether a university or research facility, does not affect a city's lobbying behaviour to a statistically significant degree.

The adjusted R-squared of the most comprehensive model is 0.46, suggesting that approximately half the variation in the outcome variable is explained by the independent variables. However, it is worth noting that this is disproportionately inflated by the addition of state dummies rather than the predictors, as the adjusted R-squared for Model 2 is only 0.20.

To examine the interaction terms, I plot the conditional marginal effects of city population at different levels of interacting variables. Figure 4 visually demonstrates the results shown in the table of coefficients. If the 95% confidence interval band overlaps zero, the relationship is not statistically significant, as zero is a plausible value for the coefficient. The plots are based on Model 4; as such, the confidence bands are estimated with HAC standard errors.

Figure 4. Conditional marginal effects plots of city population.



For the interaction between population and revenue, the relationship is not statistically significant. This suggests that the data does not provide strong evidence that the relationship between population and lobbying activity is different in cities at different levels of revenue. The same is true of the interaction with ethnic heterogeneity.

The interaction between household poverty rates and city-wide income inequality is statistically significant above a household poverty rate of approximately 14%. This suggests that above this level of household poverty, the city's population has a statistically significant and increasingly positive association with likelihood of lobbying. However, the effect is very close to zero throughout, suggesting that this relationship is positive but weak in magnitude.

The interaction between income inequality and city population is positive and significant at the 95% level for the entire range of Gini indices found in the sample. Substantively, this suggests that the effect of a larger population on lobbying is stronger when a city also has higher levels of inequality. A possible hypothetical reason for this is that cities might face more pressure to provide public goods and services if a larger proportion of the citizens are low-income or dependent on these public goods in some way. Cities may then turn to lobbying in order to secure funding or resources to meet this demand.

4. Non-Linear Models: Logistic Regressions

Another method of examining a binary outcome variable is through a logistic (Logit) regression. It models the conditional probability of the outcome with a cumulative density function. Helpfully, it constrains the predicted values to between zero and one, which a LPM is unable to do. However, the coefficients returned are log-odds ratios, which are relatively difficult to interpret. For example, the estimated coefficient on a

Table 3: Logit log-odds ratios on lobbying activity.

	Estimate	Std. Error	z value	Pr(>|z|)
(Intercept)	-15.860	0.404	-39.262	0.000
Log of City Population	0.921	0.052	17.732	0.000
Log of City Revenue	0.148	0.040	3.730	0.000
Research Facility	0.460	0.123	3.732	0.000
Public R1 University	0.022	0.070	0.316	0.752
Intergovernmental Revenue Share	-2.139	0.166	-12.854	0.000
Poverty Rate (Household)	0.013	0.004	3.582	0.000
Ethnic Heterogeneity Index	1.229	0.137	8.975	0.000
Income Inequality (Gini)	5.610	0.503	11.144	0.000
State-City Ideological Difference	-0.535	0.101	-5.286	0.000
Direct Flight to DC	0.214	0.073	2.916	0.004

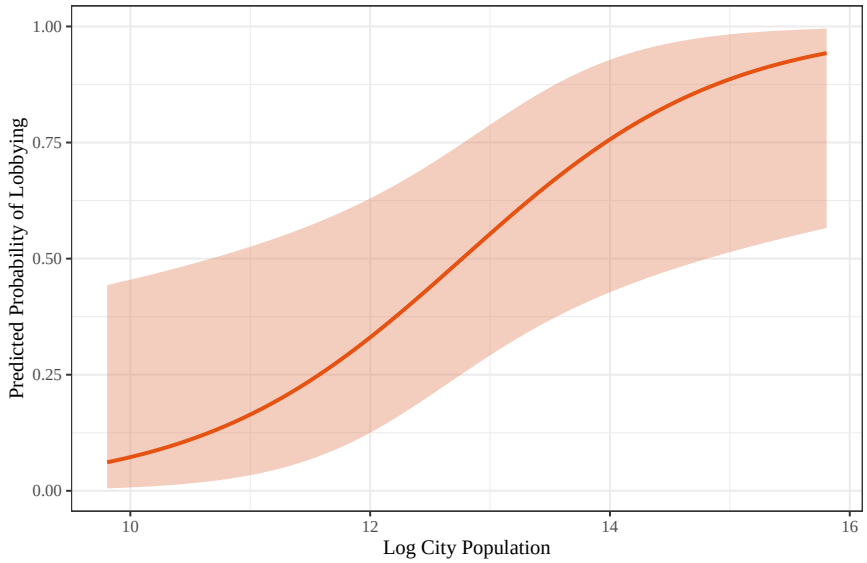
research facility is 0.46. We take the log of this value to return the multiplier on the odds of lobbying when a city goes from having no research facility to having one, *ceteris paribus*. In this case, the presence of a research facility multiplies the odds of a city lobbying by approximately 1.58. This result is statistically significant, with a very small p-value.

These interpretations are fairly unintuitive. I have therefore opted to plot the logit curves, which display the change in predicted probability of the outcome (lobbying) at different levels of the independent variable.

4.1. Logit Models: Results

The plot below displays the predicted probabilities of lobbying at different levels of city population, with 95% confidence intervals.

Figure 5. Logistic regression on city population.



The regression line has a positive slope, which corroborates the coefficients from my linear models. An increase in city population is associated with a higher predicted probability of lobbying. However, the confidence interval is wide, spanning almost half the y-axis. A city with a high population relative to my sample has anywhere from a 50% to 100% probability of lobbying, which tells us that it is more likely than

not to lobby. Next, I investigated the effects of the presence of a research facility or a public R1 university on this relationship.

Figure 6. Conditional logit: city population by federal research facility.

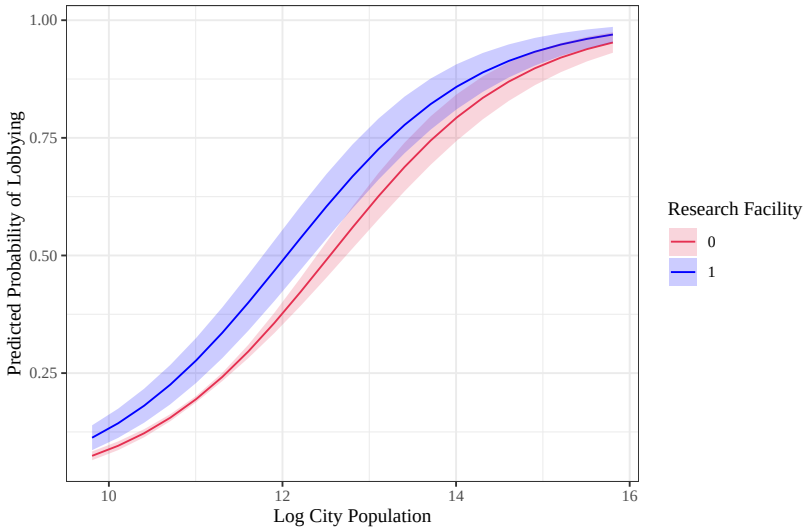
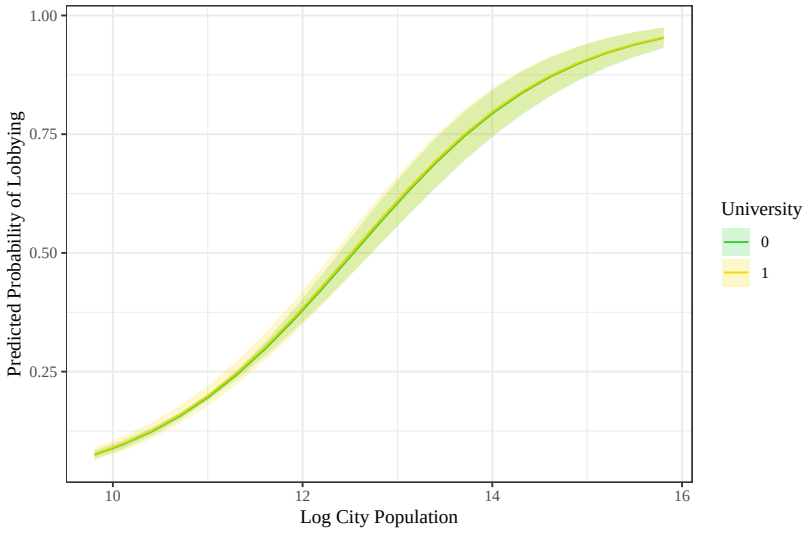


Figure 7. Conditional logit: city population by presence of public R1 university.



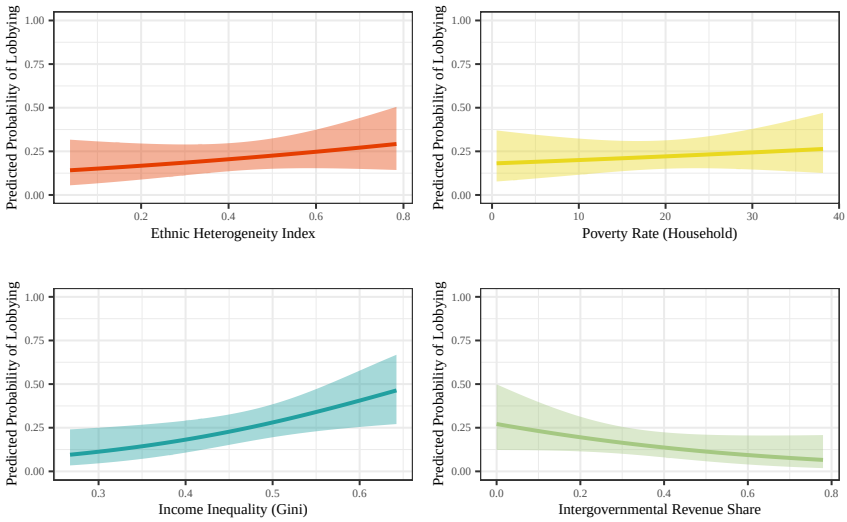
In Figure 6, we can see that the presence of a research facility creates a statistically significant difference in the predicted probability of lobbying up until a population of approximately 730k is reached, and the confidence intervals overlap. This suggests that the presence of a research facility matters more for smaller cities than for larger ones, possibly because a larger share of local employment or economy depends on these large federal institutions.

In Figure 7, the two confidence intervals overlap for the entirety of the line. As such, with this model I fail to reject the null hypothesis. The presence of a public R1 university has no statistically significant impact on the effect of city population on probability of lobbying.

Additionally, I plotted the logit predicted probability along other variables. Visually, as ethnic heterogeneity or household poverty increase, the predicted probability of lobbying increases slightly. Although the relationship is positive, the magnitude is low as the slopes are weak. Income inequality appears to have the strongest

non-linear effect on the predicted probability, as the slope rises at an increasing rate along levels of income inequality. The most unequal cities have a disproportionately higher probability of lobbying. Finally, the relationship between intergovernmental revenue share and probability of lobbying is surprisingly negative. I originally hypothesized that cities who are more dependent on other governments for revenue may be more likely to lobby. However, the results suggest that as the share of revenue increases, cities are actually less likely to lobby. This may be because cities that are highly dependent on intergovernmental revenue may have pre-established relationships with other levels of government that may preclude them from formally lobbying.

Figure 8. Logistic regression lines for additional variables.



4.2 Logit Models: Diagnostics

Non-linear logit models are unable to account for dummy fixed effects, as they induce the incidental parameters problem (Neyman and Scott 1948). As such, it is impossible to control for state fixed effects or within-state clustering in my logit regressions. Furthermore, a small sample size for institutions may decrease the validity of these results. Of the 1,262 cities in the sample, only 47 (less than 4%) have a major research facility. Therefore, my results for the conditional logit on research facilities, though statistically significant, may not be very reliable.

5. Limitations

In this section, I address the origins of the data and its limitations. The original researchers chose to narrow the scope of the study to cities with populations over 25,000 due to the availability of data through the American Community Survey (Goldstein and You 2017). They note that these cities make up over 70% of lobbying by local governments. However, even though this is the majority, it leaves a nontrivial amount of lobbying activity unaccounted for. If there are systemic patterns in lobbying by smaller cities with populations below 25,000, the data is unable to measure them. Furthermore, they note that there were 189 cities that met the population threshold but were not included because their municipal governments did not have the ability to collect and disburse their finances by themselves. As such, their data was not available through the Census of Governments (You and Goldstein 2017). Similarly, if there were patterns between the cities that were not included in the dataset, it would skew the sample and remove important information.

Lobbying activity was collected through reports submitted under the Lobbying Disclosure Act. At the time of the study, the LDA only required interest groups to file a report if the total amount paid to lobbyists exceeded $10,000; it has since been raised to $14,000 (Office of the Clerk, U.S. House of Representatives 2021). It is therefore likely that although the records made public under the LDA are the best information available, they do not capture the full scope of lobbying activity.

Finally, there are some temporal problems with the data. First, city populations grow continuously. The researchers do not mention how they account for the inevitability of cities surpassing the 25,000 population threshold over the course of the study period. Second, although the data spans 1999-2012, I was unable to obtain historical information on federal institutions from that time period. If a facility was newly constructed between then and now, it may lead to minor inaccuracies in my indicators.

6. Conclusion

The dynamics of intergovernmental lobbying, though highly salient, remain mostly understudied. This paper has attempted to shed some more light on what factors motivate a city to lobby the federal government for revenue re-allocation or other issues that may concern them. I have broadly found that population is a relatively salient predictor, as well as income inequality; however, results depend on model specifications. Additionally, I tested original indicators for federal institutions to see if these influence a city's likelihood of lobbying. I have run linear probability models and logit regressions to determine the relative importance of these factors, and highlighted some limitations of both the models and the original data.

Future research into this issue could investigate whether or not this lobbying, when it happens, is successful in achieving a city's aims. Furthermore, as the data in this study is from over a decade ago, our understanding of intergovernmental lobbying dynamics could benefit from more recent and up-to-date analysis.

Citations

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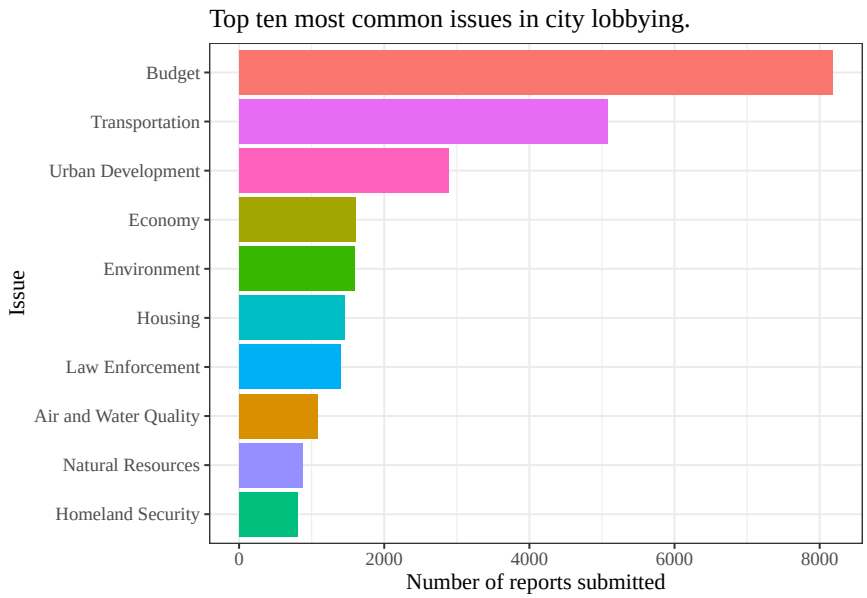
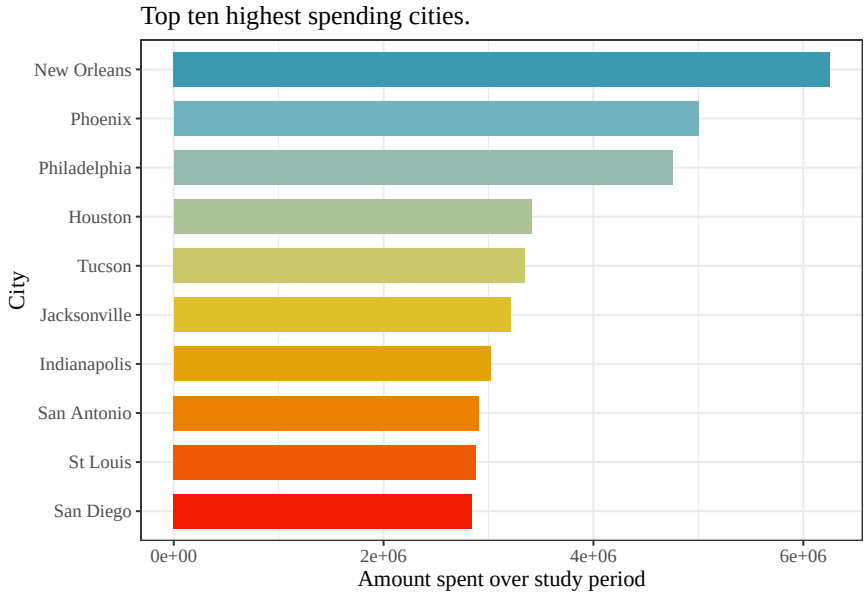
7. Appendix

7.1. Top Issues and Top Spenders